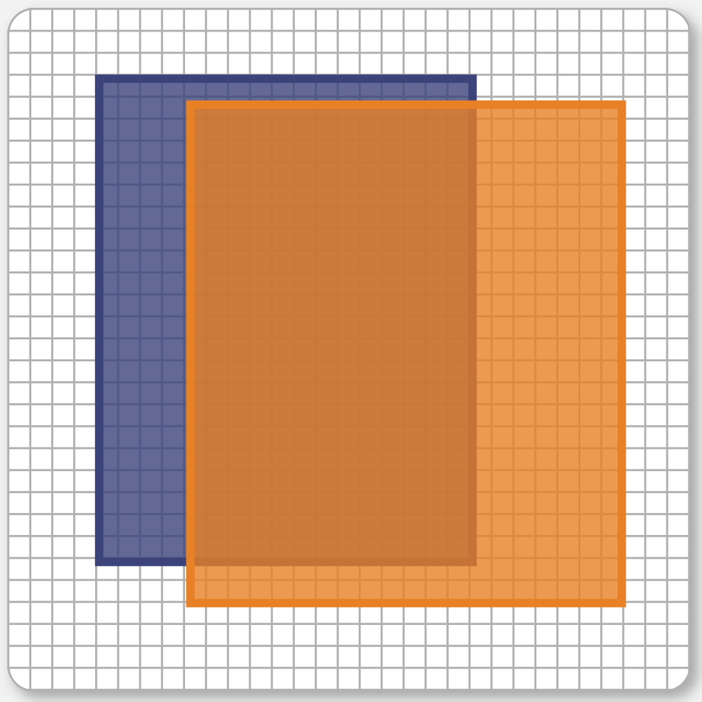
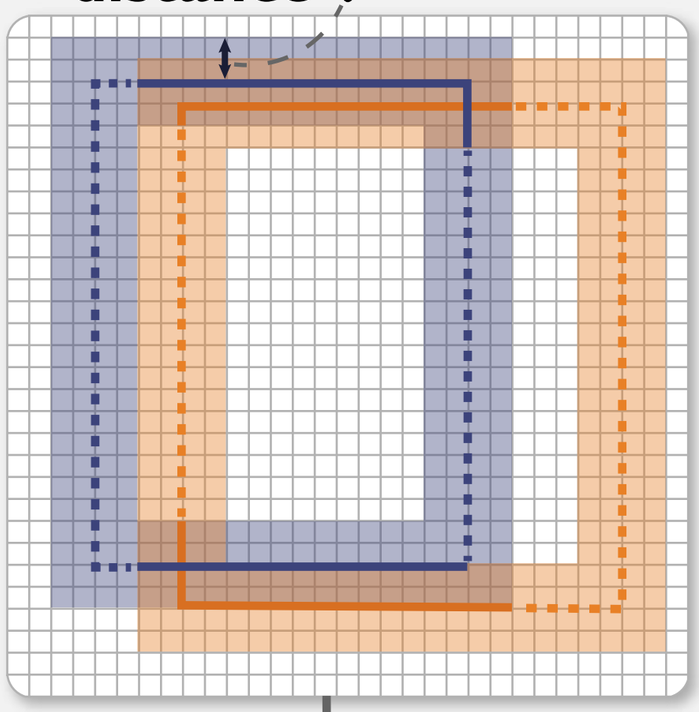
(a)



Maximum tolerated distance τ



A

Boundary of A, S_A

Border regions of A, $\mathcal{B}_A^{(\tau)}$

Boundary outside of $(S_A \cap \mathcal{B}_B^{(\tau)})$

B

Boundary of B, S_B

Border regions of B, $\mathcal{B}_B^{(\tau)}$

Boundary outside of $(S_B \cap \mathcal{B}_A^{(\tau)})$

Normalized Surface Distance (NSD)

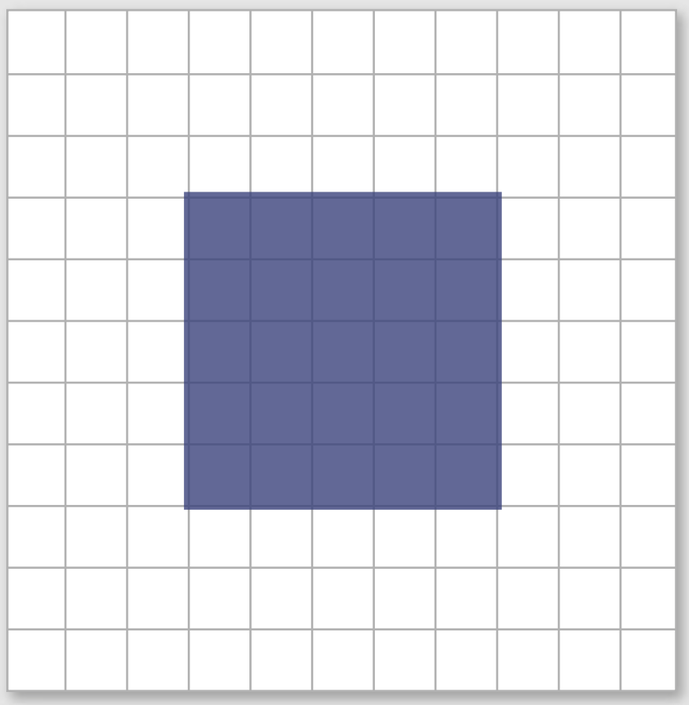
$$NSD(A,B)^{(\tau)} = \frac{\square + \square}{\square + \square}$$

$$= \frac{|S_A \cap \mathcal{B}_B^{(\tau)}| + |S_B \cap \mathcal{B}_A^{(\tau)}|}{|S_A| + |S_B|}$$

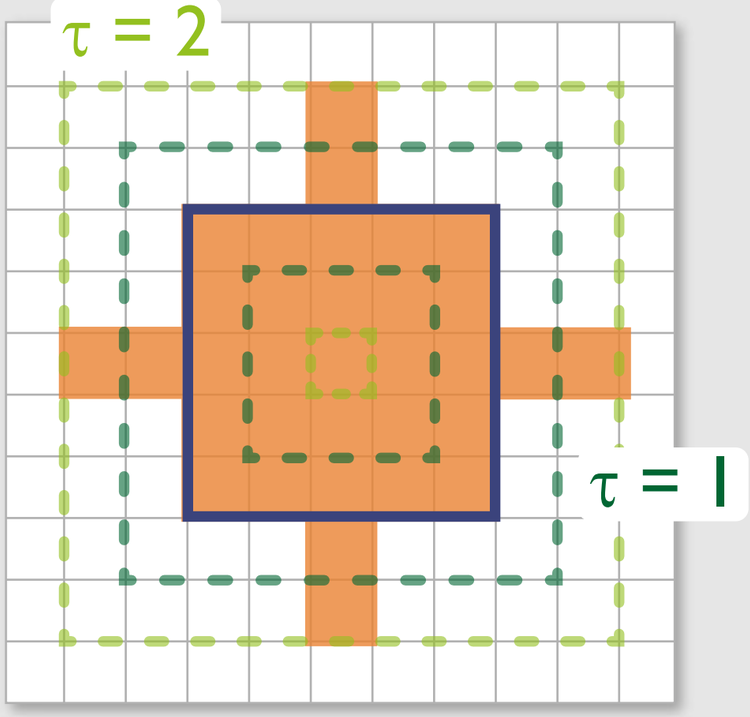
Synonyms: *Surface Dice, Normalized Surface Distance*

(b)

Reference



Prediction



***DSC* = 0.9**

IoU = 0.8

HD = 2.0

HD95 = 2.0

ASSD = 0.4

MASD = 0.4

***NSD* = 0.9, τ = 1**

***NSD* = 1.0, τ = 2**